

Definition: Random Definitions

- $\text{pr}_i: X_1 \times \dots \times X_n \rightarrow X_i, (x_1, \dots, x_n) \mapsto x_i$
- The **power set** of X is $\mathcal{P}(X) = \{U: U \subseteq X\}$
- A subset of $\mathcal{P}$ is a **system of subsets** of X .
- $\emptyset$ is understood to be the linear combination of no vectors.
- A (an operator/mat/etc) is **nilpotent** with **degree** k if $A^k = 0, k \in \mathbb{N}$.
- For a set X and $f: X \rightarrow X, Y \subseteq X$ is **stable under** f if $f(Y) \subseteq Y$.

Theorem: Random Theorems

- Group axioms for $(G, *)$: G closed under $*$; $*$ associative $[g*(h*k)] = (g*h)*k$, $\exists$ unique identity $e \in G$; $\exists$ unique inverse $g^{-1} \in G \forall g$.
- For G, H grps, $f: G \rightarrow H$ is a group hom. if $f(ab) = (f(a))f(b)$.
- For f a group hom. $\ker(f) = \{a \in G: f(a) = e\}$ is a normal subgrp of G .
- O-S: For G a group acting on a finite set X , $|\text{Orb}(x)| = |G|/|\text{Stab}(x)|$
- Lagrange's Theorem: For G a finite group and $H \leq G$ a normal subgroup (invariant under conjugation, $ghg^{-1} \in H \forall g \in G, h \in H$); $|G| = |H| |G/H|$

Remark: Notation

- VS = Vector Space. VsubS = Vector subSpace. F -VS = F -Vector Space
- 'lin-ind' = 'linearly independent' and 'lin-dep' = 'linearly dependent'
- 'almost all' means 'all but finitely many'
- 'n-z' = 'non-zero'

Vector Spaces

Definition: Systems Of Linear Equations

Let F be a field. We seek to solve the system of n linear equations in m unknowns, described more easily by the extended coefficient matrix.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

- The set $L \subseteq F^m$ of m -tuples satisfying the equations is the **solution set**.
- For a lin sys with soln set $\neq \emptyset$, all solns are obtained by adding an arbitrary soln of the homogenised sys to a fixed soln of the sys componentwise.

Major Definition: Fields And Vector Spaces

- A **field** F is a set with functions $+$: $F \times F \rightarrow F; (\lambda, \mu) \mapsto \lambda + \mu$ and $\cdot$: $F \times F \rightarrow F; (\lambda, \mu) \mapsto \lambda \mu$ s.t. $(F, +)$ and $(F, \cdot)$ are abelian groups with respective identities $0_F, 1_F$, and $\lambda(\mu + \nu) = \lambda\mu + \lambda\nu \in F$.
- $0\vec{v} = \vec{0}, (-1)\vec{v} = -\vec{v}$
- $\forall \lambda \in F, \lambda \vec{0} = \vec{0}$
- If $\lambda \vec{v} = \vec{0}, \lambda = 0$ or $\vec{v} = \vec{0}$
- For X a set and V an F -VS, the set $\text{Maps}(X, V)$ of mappings $X \rightarrow V$ is an F -VS (define $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda(f(x))$).
- Given F -VSs $V_1, \dots, V_n, V_1 \times \dots \times V_n$ is an F -VS if we define addition/mult by scalar componentwise. We denote this VS by $V_1 \oplus \dots \oplus V_n$ and name it the **external direct sum**. $F^n = F \oplus \dots \oplus F$ n times.
- Two VsubSs V_1, V_2 of a VS V are **complementary** if addition defines a bijection: $V_1 \times V_2 \rightarrow V$
- A nice example of an infinite-dim VS is the space of all sequences (ie working over $\mathbb{R}$, real seqs).
- A nice endomorphism with this VS is 'shifting' all the indices of the seq.
- For a set X , the set $\text{Maps}(X, F)$ equipped with pointwise addition and scalar multiplication is an F -VS.

Definition: Vector Subspaces

- Let U be a subset of a VS V . U is a **vector subspace** if $\vec{0} \in U$ and $\forall \vec{u}, \vec{v} \in U, \lambda \in F: \vec{u} + \vec{v} \in U$ and $\lambda \vec{u} \in U$.
- The solution set of a lin sys of eqns seen earlier is a vector space
- $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n, n \in \mathbb{N}$ is a **linear combination** of $\vec{v}_1, \dots, \vec{v}_n$.
- A subset is a **hyperplane** of a VS if it is a proper VsubS (a VsubS $\neq V$) and together with one other $\vec{v}$ generates the whole VS.
- The intersection of two VSs is itself a VS

Definition: The Free Vector Space

The **free vector space** on $X, F(X) \subseteq \text{Maps}(X, F)$, is the VsubS of all mappings which send almost all $x \in X$ to zero.

Definition: Span and Generating Sets

- Let T be a subset of an F -VS V . $\langle T \rangle$, the **vector subspace generated by T** or **span** of T is the set of all linear combinations of vectors in T .
- $\langle \emptyset \rangle = \vec{0}$ If $v \in T, \langle T \cup \{v\} \rangle = \langle T \rangle$.
- A subset of V is a **generating/spanning set** if its span is V .
- A VS with a finite generating set is **finitely generated**.
- Given $V_1, \dots, V_n$ VsubSs of V , we define the **sum** $V_1 + \dots + V_n$ to be the subspace $(V_1 \cup \dots \cup V_n)$ of V .
- If the hom. given by addition: $V_1 \times \dots \times V_n$ is also an injection the sum is **direct** and can be written $V_1 \oplus \dots \oplus V_n$.

Definition: Linear Independence

- A subset L of a VS V is **linearly independent** if for all $\alpha_1, \dots, \alpha_n \in F$ and distinct $\vec{v}_1, \dots, \vec{v}_n \in L, \alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ only if all $\alpha_i = 0$. Otherwise it is **linearly dependent**.
- The FEILA: For VS $V, L \subseteq V$ lin-ind, and $E \subseteq V$ generating, $|L| \leq |E|$.

Definition: Bases

- Let V be an F -VS. A basis of V is a lin-ind generating set in V .
- The family $(\vec{v}_i)_{1 \leq i \leq n}$ is a basis of V iff the 'evaluation mappings' $\Phi: F^n \rightarrow V; (\alpha_1, \dots, \alpha_n) \mapsto (\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n)$ is a bijection.
- TFAE for a subset E of a VS V :
 - (i) E is a basis
 - (ii) E is minimal among generating sets ($\forall \vec{v} \in E, E \setminus \{\vec{v}\}$ is not generating)
 - (iii) E is maximal among lin-ind subsets ($\forall \vec{v} \in E, E \cup \{\vec{v}\}$ is lin-dep)
- A family $(\vec{v}_i)_{i \in I}$ is a basis of V iff for all $\vec{v} \in V$ there exists exactly one family $(\alpha_i)_{i \in I}$ of $\alpha_i \in F$ s.t. almost all $\alpha_i = 0$, and $\vec{v} = \sum_{i \in I} \alpha_i \vec{v}_i$.

Major Theorem: Cardinality Of Bases

Let V be a finite VS. Then V has a finite basis, cannot have an infinite basis, and any two bases of V have the same cardinality

Lemma: The Exchange Lemma, Steinitz

- Let V be a VS, $E \subseteq V$ generating, $L \subseteq V$ and $M \subseteq E$ lin-ind, L finite:
 - The Exchange Lemma: If $\vec{w} \in V \setminus M$ is s.t. $M \cup \{\vec{w}\}$ is lin-ind, then there exists $\vec{z} \in E \setminus M$ s.t. $(E \setminus \{\vec{z}\}) \cup \{\vec{w}\}$ is generating for V .
 - Steinitz Exchange Theorem: There exists an injection $\phi: L \hookrightarrow E$ s.t. $(E \setminus \phi(L)) \cup L$ is generating for V (apply Exchange Lemma repeatedly)

Definition: Dimensions Of Vector Spaces

- Let V be a finitely-generated VS, $L \subseteq V$ lin-ind, $E \subseteq V$ generating. For a finitely-generated VS V , its **dimension dim** is the cardinality of any basis of V . For dimension as an F -VS write $\dim_F V$. Also:
 - $|L| \leq \dim V$, if $|L| = \dim V$ then L is a basis
 - $|E| \geq \dim V$, if $|E| = \dim V$ then E is a basis
 - A proper VsubS of V has a strictly smaller dimension than V
 - For U, W subspaces of $V, \dim(U+W) + \dim(U \cap W) = \dim(U) + \dim(W)$.
 - Given $V_1, \dots, V_n$ F -VSs, $\dim(V_1 \oplus \dots \oplus V_n) = \dim(V_1) + \dots + \dim(V_n)$

MDfn: Linear Mappings: Homo/Iso/Endo/Automorphisms

- Let V, W be F -VSs, $B \subseteq V$ be a basis. A mapping $f: V \rightarrow W$ is **(F-)linear** or a **homomorphism of F-VSs** if $\forall \vec{v}_1, \vec{v}_2 \in V$ and $\lambda \in F: f(\vec{v}_1 + \vec{v}_2) = f(\vec{v}_1) + f(\vec{v}_2)$ and $f(\lambda \vec{v}_1) = \lambda f(\vec{v}_1)$.
- An **isomorphism** is a bijective linear mapping. Isomorphisms are invertible, and the inverse mapping is also an isomorphism
- An **endomorphism** is a homomorphism from a VS to itself
- An **automorphism** is an isomorphism from a VS to itself.

Major Theorem: Properties Of Morphisms

- Let V, W be F -VSs, $B \subseteq V$ be a basis,
 - The set of all homs.
 - $V \rightarrow W$ is $\text{Hom}_F(V, W) = \text{Hom}(V, W) \subseteq \text{Maps}(V, W)$, and is itself a VS
 - The auto.s of a V form a subgroup of its permutation group, the **general linear** or **automorphism** group $GL(V) = \text{Aut}(V) \subseteq \text{Maps}^\times(V, V)$.
 - If $f: V \rightarrow W$ and $g: U \rightarrow V$ are homs., $(f \circ g): U \rightarrow W$ is a hom.
 - The image and preimage of a VS under a linear mapping are VSs.
 - A hom. can be restricted to a bijec $\text{Hom}_F(V, W) \rightarrow \text{Maps}(B, W), f \mapsto f|_B$, ie a hom. determines/is determined by the values it takes on a basis.
 - Every injective hom. $f: V \rightarrow W$ has a **left inverse** g s.t. $g \circ f = \text{id}_V$
 - Every surjective hom. $f: V \rightarrow W$ has a **right inverse** g s.t. $f \circ g = \text{id}_W$
 - For $n \geq 0 \in \mathbb{Z}$, a VS V over F is isomorphic to F^n iff $\dim V = n$.
 - A point sent to itself by a hom. is a **fixed point** of the mapping. Given a mapping $f: X \rightarrow X$, the set of fixed points is $X^f = \{x \in X: f(x) = x\}$.
 - Given VS V , VsubSs $V_1, \dots, V_n$, and linmaps $f_i: V_i \rightarrow V; f: V_1 \oplus \dots \oplus V_n \rightarrow V, f(v_1, \dots, v_n) = f_1(v_1) + \dots + f_n(v_n)$ is an isomorphism. We then write $V = V_1 \oplus V_2$ and say V is the **internal direct sum** of V_1, V_2 .

Definition: Image and Kernel

- Let V, W be VSs and $f: V \rightarrow W$ be a hom.
 - The **image** of f , $\text{im}(f) = F(V) \subseteq W$, is a VsubS of W .
 - The **kernel** of f , $\ker(f) := f^{-1}(\vec{0}) = \{\vec{v} \in V: f(\vec{v}) = \vec{0}\}$, is a VsubS of V .
 - A linear mapping f is injective iff its kernel is $\vec{0}$.
 - The image of a VsubS under a linear mapping is again a VsubS, and the preimage of a VsubS under a linear mapping is again a VsubS.

Major Theorem: Rank-Nullity Theorem

- Let V, W be VSs and $f: V \rightarrow W$ a linear mapping. Then $\dim V = \dim(\ker(f)) + \dim(\text{im}(f))$. $\dim(\text{im}(f))/\dim(\ker(f))$ are the **rank/nullity** of f .

Linear Mappings and Matrices

Definition: Matrices

- A: $\{1, \dots, n\} \times \{1, \dots, m\} \rightarrow Z$ is an $n \times m$ -matrix with coefficients in Z .
- The set of all $n \times m$ -matrices with coefficients in Z is denoted $\text{Mat}(n \times m; Z) := \text{Maps}(1, \dots, n \times 1, \dots, m, Z)$
- If $m = n$ we have a **square matrix** and write $\text{Mat}(n, Z)$
- For square matrix A , its **trace** $\text{tr}(A)$ is the sum of its diagonal entries
- $P \in \text{Mat}(n, \mathbb{R})$ is **orthogonal** if $P^T P = I_n$, ie $P^T = P^{-1}$.
- $P \in \text{Mat}(n, \mathbb{C})$ is **unitary** if $\overline{P}^T P = I_n$, ie $\overline{P}^T = P^{-1}$.
- $P \in \text{Mat}(n, \mathbb{C})$ is **hermitian** if $P = \overline{P}^T$, and the mapping on the standard $\mathbb{R}^n$ described by P is self-adjoint.
- For $A \in \text{Mat}(n, \mathbb{C}), \exp(A) := \sum_{k=0}^{\infty} \frac{1}{k!} A^k$.
- For commutative $A, B, \exp(A+B) = \exp(A)\exp(B)$

Theorem: Linking Linear Mappings And Matrices

Let F be a field and $m, n \in \mathbb{N}$. Then is a bijection M between the space of homs $F^m \rightarrow F^n$ and $\text{Mat}(n \times m; F)$ which maps $f \mapsto [f]$. $M(f) := [f]$ is the **representing matrix** of f , whose columns are the images under f of the standard basis elements of F^m , ie $[f] = (f(\vec{e}_1)) \dots (f(\vec{e}_m))$.

Definition: Composing Mappings vs Matrix Multiplication

- Let $g: F^\ell \rightarrow F^m$ and $f: F^m \rightarrow F^n$ be linmaps. The representing mat of their composition is the product of their representing mats: $[f \circ g] = [f][g]$.
- For $\ell, m, n \in \mathbb{N}, F$ a field, $A \in \text{Mat}(n \times m; F), B \in \text{Mat}(m \times \ell; F); AB \in \text{Mat}(n \times \ell)$ is defined by $(AB)_{ik} = \sum_{j=1}^m A_{ij} B_{jk}$.

Definition: The General Linear Group

The set of invertible $n \times n$ matrices in a field F form a group under matrix multiplication, called the **General Linear group** of $(n \times n)$ matrices and written $GL(n; F) := \text{Mat}(n; F)^\times$

Definition: Properties Of Matrices

- If $A, A' \in \text{Mat}(n \times m; F), B, B' \in \text{Mat}(m \times \ell; F), C \in \text{Mat}(\ell \times k; F)$, then:
 - (i) $(A+A')B = AB + A'B$
 - (ii) $A(B+B') = AB + AB'$
 - (iii) $(AB)C = A(BC)$
- A square A is **invertible** if $\exists A^{-1}$ of same size s.t $AA^{-1} = A^{-1}A = I$
- An **elementary** matrix is one which differs from I in at most one entry
- Every $M \in \text{Mat}(n, F)$ can be written as a product of elementary matrices
- A matrix whose only non-zero entries lie on the diagonal, with first ones along the diagonal then zeroes, is in **Smith Normal Form**.
- For each $A \in \text{Mat}(n \times m; F), \exists$ invertible P, Q s.t. PAQ is in SNF.
- The **rank** of a matrix A is the dimension of the subspace of F^n generated by the columns or rows of A (the dim of each are always equal).
- A is of **full rank** if the rank is equal to the number of rows or columns
- $\text{rk}(A+B) \leq \text{rk} A + \text{rk} B$, and $\text{rk}(AB) \leq \min(\text{rk} A, \text{rk} B)$.

Definition: Representing Matrices W.R.T Bases

- For F a field, V, W F -VSs with ordered bases $A = (\vec{v}_1, \dots, \vec{v}_m), B = (\vec{w}_1, \dots, \vec{w}_n)$, to each hom. $F: V \rightarrow W$ we associate a **representing matrix** $\mathcal{B}[f]_A$ with entries a_{ij} defined by $f(\vec{v}_j) = a_{1j} \vec{w}_1 + \dots + a_{nj} \vec{w}_n \in W$.
- Then $M_{\mathcal{B}}^{\mathcal{A}}: \text{Hom}_F(V, W) \rightarrow \text{Mat}(n \times m; F), f \mapsto \mathcal{B}[f]_A$ is an iso.
- We call $M_{\mathcal{B}}^{\mathcal{A}}[f] = \mathcal{B}[f]_A$ the **representing matrix** of f with respect to the bases A and B . The columns of $\mathcal{B}[f]_A$ give the coords of the images of the basis vectors from A wrt the basis B .
- Let U, V, W be finite-dimensional F -VSs with ordered bases A, B, C , and $f: U \rightarrow V, g: V \rightarrow W$ be linmaps. Then the representing matrix of $(g \circ f): U \rightarrow W$ is the matrix product of the representing matrices of f and g : $\mathcal{C}[g \circ f]_A = \mathcal{C}[g]_B \mathcal{B}[f]_A$
- Let A an ordered basis of a finite-dim VS V , the inverse of the 'evaluation mapping' $\Phi_A: F^m \rightarrow V, (\alpha_1, \dots, \alpha_m)^T \mapsto \alpha_1 \vec{v}_1 + \dots + \alpha_m \vec{v}_m$ is given by $\vec{v} \mapsto A[\vec{v}]$. $A[\vec{v}]$ is the **representation** of $\vec{v}$ w.r.t. the basis A .
- Let A, B be ordered bases of finite-dim VSs V, W and $f: V \rightarrow W$ be a linmap. Then for $\vec{v} \in V, \mathcal{B}[f(\vec{v})] = \mathcal{B}[f]_A \mathcal{A}[\vec{v}]$

Definition: Change Of Matrix By Change Of Basis

- Let $A = (\vec{v}_1, \dots, \vec{v}_n)$ and $B = (\vec{w}_1, \dots, \vec{w}_n)$ both be ordered bases of an F -VS V . Then the matrix representing the identity map w.r.t. these bases, $\mathcal{B}[\text{id}_V]_A$ is called a **change of basis matrix**. By definition its entries are given by the equalities $\vec{v}_j = \sum_{i=1}^n a_{ij} \vec{w}_i$.
- Let V, W be finite-dim F -VSs and $f: V \rightarrow W$ be a linmap. Suppose A, A' and B, B' are ordered bases of V and W respectively. Then $\mathcal{B}'[f]_{A'} = \mathcal{B}'[\text{id}_W] \mathcal{B}[\mathcal{B}[f]_A \mathcal{A}[\text{id}_V]_{A'}]$.
- For A, A' ordered bases of finite-dim VS V , and f an endo. of $V: A'[\mathcal{B}[f]_{A'} = A[\mathcal{B}[f]_A \mathcal{A}[\text{id}_V]_{A'}]$
- $N = \mathcal{B}[f]_B, M = A[f]_A$ are similar if $N = T^{-1}MT$ ($T := A[\text{id}_V]_B$).
- Let $f: V \rightarrow W$ be a linmap between finite-dim F -VSs. There exists ordered bases A of V and B of W s.t. $\mathcal{B}[f]_A$ is in Smith Normal Form

Rings And Modules

Major Definition: A Ring

- A **ring** is a set R equipped with two operations $+, \cdot$ such that:
 - (i) $(R, +)$ is an abelian group
 - (ii) $(R, \cdot)$ is a monoid (satisfies associativity and identity)
 - (iii) $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$ and $(a+b) \cdot c = (a \cdot c) + (b \cdot c)$ (distributivity).
- The element $1 \in R$ is uniquely defined as the identity of $(R, \cdot)$.
- A ring in which $\cdot$ is commutative is a **commutative ring**.
- The **null/zero ring** is the ring where R is a single element set.
- The integers mod m (or $\mathbb{Z}/m\mathbb{Z}$) is a ring made up of congruence classes.
- A field is a non-zero commutative ring
- F where every $n \neq 0$ $a \in F$ has inverse $a^{-1} \in F$, ie $a \cdot a^{-1} = a^{-1} \cdot a = 1$
- For $m \in \mathbb{N}, \mathbb{Z}/m\mathbb{Z}$ is a field iff m is prime
- For $a, b \in \mathbb{R}, 0a = 0 = a0; (-a)b = -(ab) = a(-b)$; and $(-a)(-b) = ab$.
- For $m \in \mathbb{Z}$, the m -th multiple ma of $a \in$ **abelian** R is $ma = a + \dots + a$ m times if $m > 0, 0a = 0$. If $m < 0, (-m)(a) = -(ma)$.
- For $m, n \in \mathbb{Z}, a, b \in R: m(a+b) = ma + mb; (m+n)a = ma + na; m(na) = (mn)a$; and $m(ab) = (ma)b = a(mb)$.

Major Definition: Units, Zero-Divisors, Integral Domains

- Let R be a ring, and $a \in R$. a is a **unit** if it is invertible in R , ie there exists $aa^{-1} = 1 \in R$ s.t. $aa^{-1} = a^{-1}a = 1$.
- A $n \neq 0$ a is a **zero-divisor** if $\exists n \neq 0 b \in R$ s.t. either $ab = 0$ or $ba = 0$.
- An **integral domain** is a $n \neq 0$ commutative ring with no zero-divisors.
- The set of units of a ring, $R^\times$, form a group under multiplication.
- For $m \in \mathbb{N}, \mathbb{Z}/m\mathbb{Z}$ is an integral domain iff m is prime
- Every finite integral domain is a field
- Let $a, b, c \in R$, an int-dom. If $ab = ac$ and $a \neq 0$, then $b = c$.

Definition: Polynomials

- Let R be a ring. A **polynomial over R** is an expression of the form $P = a_m x^m + \dots + a_2 x^2 + a_1 x + a_0$ for $0 \leq m \in \mathbb{Z}$ and $a_i \in R$.
- The set of all polynomials over R is denoted $R[x]$.
- When the leading coefficient is one, the polynomial is **monic**
- Defined with the usual addition/multiplication rules for polynomials, $R[x]$ becomes a ring, called the **ring of polynomials with coefficients in R** . The zero/identity of R are the zero/identity of $R[x]$.
- If R is commutative/an int-dom/has no zero-divisors, so is or does $R[x]$.
- If R has no zero-divisors, $\deg(PQ) = \deg(P) + \deg(Q)$ for $n \neq P, Q \in R[x]$.
- Div&Rem Thm: Let R be an int-dom and $P, Q \in R[x], Q$ monic. Then $\exists$ unique $A, B \in R[x]$ s.t. $P = AQ + B$; $\deg(Q) > \deg(B)$ or $B = 0$
- For R a commutative ring $P \in R[x]$ can be **evaluated** at $\lambda \in R$ to give $P(\lambda)$. In this way we have a mapping $R[x] \rightarrow \text{Maps}(R, R)$.
- $\lambda \in R$ is a root of P if $P(\lambda) = 0$, and iff $(x - \lambda)$ divides $P(x)$.
- For R an int-dom, $n \neq P \in R[x] \setminus \{0\}$ has at most $\deg(P)$ roots in R .

Definition: Algebraic Closure

- A field F is **algebraically closed** if each non-constant polynomial $P \in F[x] \setminus F$ with coefficients in F has a root in F .
- If F alg-clsd, all $n \neq P \in F[x] \setminus \{0\}$ decompose uniquely, up to reordering, into linear factors: $P = c(x - \lambda_1) \dots (x - \lambda_n); n \geq 0, c \in F^\times, \text{ all } \lambda_i \in F$.

Major Definition: Ring Homomorphisms

- Let R, S be rings. $f: R \rightarrow S$ is a **ring homomorphism** if $\forall x, y \in R$:
 - $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$
 - For $x \in R, n \in \mathbb{N}, f(x^n) = (f(x))^n$
 - The zero ring homomorphism is $f: R \rightarrow S; r \mapsto 0$.
 - Isomorphisms etc are akin to linear mappings
 - $f(1_R)$ is not necessarily equal to 1_S .
 - If S has no zero-divs then either f is the zero hom., or $f(1_R) = 1_S$.
 - The composition of hom.s is a hom. and the inverse of an iso. is an iso.
 - If $f: R \rightarrow S$ a ring hom., then $\forall x, y \in R$ and $m \in \mathbb{Z}$:
 - $f(0_R) = 0_S; f(-x) = -f(x); f(x-y) = f(x) - f(y); f(mx) = mf(x)$.

Major Definition: Ideals

- A subset I of a ring R is an **ideal**, written $I \leq R$, if:
 - (1) $I \neq \emptyset$
 - (2) I is closed under subtraction
 - (3) For all $i \in I$ and $r \in R$, we have $ri, ir \in I$

Thm: Universal Prop. Of The Set Of Equivalence Classes
 Let $\sim$ be an equivrel on X .
 If $f: X \rightarrow Z$ is a mapping s.t. $x \sim y \implies f(x) = f(y)$; there is a unique mapping $\bar{f}: (X/\sim) \rightarrow Z$, $\bar{f}(E(x)) := f(x)$, and $f = \bar{f} \circ \text{can}$.
 Often when defining a map $g: (X/\sim) \rightarrow Z$, we define $f: X \rightarrow Z$ and say g is **well-defined** to indicate that $x \sim y \implies f(x) = f(y)$
 as this guarantees the existence of $g = \bar{f}$.

Definition: Cosets and Factor Rings

Let R be a ring, $I \trianglelefteq R$ an ideal of R , and $\sim$ defined by $x \sim y \iff x - y \in I$.
 $a + I := \{a + i : i \in I\} \subseteq R$
 is a **coset** of I in R or the **coset** of x w.r.t. I in R .
 The set R/I , the **factor ring** of R by I or **quotient** of R by I , is the set $(R/\sim)$ of cosets of I in R .
 • R/I is a ring, with addition defined by $(x+I) + (y+I) = (x+y)+I$ and multiplication by $(x+I)(y+I) = xy+I$ for all $x, y \in R$.
 • If R is commutative so is R/I .
 • R/I is a nonzero ring if and only if $I \neq R$.
 • If I is proper, and $r \in R^\times$; $r+I \in (R/I)^\times$ with $(r+I)^{-1} = r^{-1}+I$.

Theorem: The Universal Property Of Factor Rings

Let R, S be rings and $I \trianglelefteq R$. Then:
 1. The mapping $\text{can}: R \rightarrow R/I, r \mapsto r+I$ is a surjective ring hom. with kernel I .
 2. For $f: R \rightarrow S$ a ring hom. with $f(I) = \{0_S\}$, ie $I \subseteq \ker f$, there exists a unique ring hom. $\bar{f}: R/I \rightarrow S$ s.t. $f = \bar{f} \circ \text{can}$.

Major Theorem: The First Isomorphism Theorem For Rings

Let R, S be rings. Then every ring hom. $f: R \rightarrow S$ induces a ring iso.: $\bar{f}: R/\ker f \xrightarrow{\sim} \text{im } f$

Definition: Modules

A (left) **module** M over a ring R or an **R -module** is a pair consisting of an abelian group $M = (M, +)$ and a mapping $R \times M \rightarrow M; (r, a) \mapsto ra$ such that for all $r, s \in R$ and $a, b \in M$ the following hold:
 • $r(a+b) = (ra) + (rb)$
 • $r(sa) = (rs)a$
 • $(r+s)a = (ra) + (sa)$
 • $1_R a = a$
 Let R be a ring and M an R -module. Then for all $r \in R, a \in M$: $0_R a = 0_M$; $r 0_M = 0_M$; $(-r)a = r(-a) = -(ra)$

Definition: Module Homomorphisms/Kernel/Image/etc

Let R be a ring and M, N be R -modules.
 A mapping $f: M \rightarrow N$ is an **(R) -homomorphism** if for all $a, b \in M$ and $r \in R$: $f(a+b) = f(a) + f(b)$, and $f(ra) = rf(a)$.
 The **kernel** of f is $\ker f = \{a \in M : f(a) = 0_N\} \subseteq M$ and the **image** of f is $\text{im } f = \{f(a) : a \in M\} \subseteq N$.
 If f is a bijection then it is an **$(R\text{-module})$ isomorphism** with $M \cong N$.
 • The composition of two R -hom.s is an R -hom.

Definition: Submodules

A $\neq \emptyset$ subset M' of an R -module M is a **submodule** if M' is an R -module w.r.t. the operations of the R -module M restricted to M' .
 Test For A Submodule:
 Let R be a ring and M be an R -module. $M' \subseteq M$ is a submodule iff:
 (1) $0_M \in M'$ (2) $a, b \in M' \implies a-b \in M'$
 (3) $r \in R$ and $a \in M' \implies ra \in M'$
 Let R be a ring, M, N be R -modules, $f: M \rightarrow N$ be an R -hom.. Then:
 • $\ker f$ is a submodule of M and $\text{im } f$ is a submodule of N .
 • f is injective if and only if $\ker f = 0_M$.
 Let R be a ring, M an R -module, and $T \subseteq M$. Then the **sub-module** of M **generated by** T is the set $\langle T \rangle = \{r_1 t_1 + \dots + r_m t_m : t_1, \dots, t_m \in T, r_1, \dots, r_m \in R\}$ together with the zero element if $T = \emptyset$. Submodules have several properties akin to subspaces/ideals:
 • M is **finitely generated** if it is generated by a finite set
 • M is called **cyclic** if it is generated by a singleton
 • $\langle T \rangle$ is the smallest submodule of M containing T , for $T \subseteq M$.
 • The intersection of any collection of submodules of M is a submodule of M .
 • If M_1, M_2 are submods of M , $M_1 + M_2 := \{a+b : a \in M_1, b \in M_2\}$ is too.

Definition: Factor Modules

Let R be a ring, M an R -module, and N a submodule of M .
 For $a \in M$, the **coset** of a w.r.t. N in M is $a+N = \{a+b : b \in N\}$.
 • It is a coset of N in the abelian group M , so is an equiv class for the equivrel $a \sim b \iff a-b \in N$.
 • **factor/quotient** of M by N is the set $(M/\sim)$ of all cosets of N in M .
 • This becomes an R -module by defining: $(a+N) + (b+N) := (a+b)+N$ and $r(a+N) = ra+N$ for all $a, b \in M, r \in R$.
 • The R -module M/N is the **factor module** of M by the submodule N .

Theorem: Universal Property Of Factor Modules

Let R be a ring, L, M be R -modules, and N a submodule of M :
 1. The mapping $\text{can}: M \rightarrow M/N, a \mapsto a+N$ for all $a \in M$ is a surjective R -hom. with kernel N .
 2. If $f: M \rightarrow L$ is an R -hom with $f(N) = \{0_L\}$ so that $N \subseteq \ker f$, there exists unique hom. $\bar{f}: M/N \rightarrow L$ such that $f = \bar{f} \circ \text{can}$.

Major Theorem: First Isomorphism Theorem For Modules
 Let R be a ring and M, N be R -modules.
 Then every R -hom. $f: M \rightarrow N$ induces an R -iso. $\bar{f}: M/\ker f \xrightarrow{\sim} \text{im } f$
 • 2^{nd} Theorem: For K, L sub-mods of M ; L is a submod of $K+L = \{b+c : b \in K, c \in L\}$ and $K \cap L$ is a submod of K . Also, $\frac{K+L}{K \cap L} \cong \frac{K}{K \cap L} \oplus \frac{L}{K \cap L}$
 • 3^{rd} Theorem: Let K, L be sub-mods of M , with $L \subseteq K$. Then K/L is a submod of M/L and $\frac{M/L}{K/L} \cong M/K$.

Determinants And Eigenvalues

Definition: Permutations And Their Signs

The group (under composition) of all permutations of the set $\{1, \dots, n\}$ to itself, denoted $\mathfrak{S}_n$, is the n -**th symmetric group**. It has order $|\mathfrak{S}_n| = n!$.
 • A **transposition** is a perm that swaps two elements (no effect on others).
 • An **inversion** of a $\sigma \in \mathfrak{S}_n$ is a pair (i, j) s.t. $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$.
 • The number of inversions of a σ is the **length** of σ , $\ell(\sigma)$.
 • The **sign** of σ is the parity of the length of σ : $\text{sgn}(\sigma) = (-1)^{\ell(\sigma)}$.
 • If $\text{sgn}(\sigma) = 1/-1$, σ is an **even/odd permutation**.
 • $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$
 • The set of even perms in $\mathfrak{S}_n$ forms a subgroup (it is the kernel of sgn of a grp hom.) of $\mathfrak{S}_n$ called the **alternating group** and denoted A_n .

Definition: Cofactors And Adjugate Matrices

Let $A \in \text{Mat}(n; R)$ for a comring $R, n, i, j \in \mathbb{N}$ s.t. $1 \leq i, j \leq n$. Then:
 • The (i, j) **cofactor** of A is $C_{ij} = (-1)^{i+j} \det(A(i, j))$ where $A(i, j)$ is the matrix obtained by deleting the i^{th} row and j^{th} column of A .
 • The **adjugate matrix** $\text{adj}(A)$ is the $(n \times n)$ matrix whose entries are $\text{adj}(A)_{ij} = C_{ji}$ (note order swap of i and j !).

Major Definition: The Determinant

Let R be a comring and $n \in \mathbb{N}$. The **determinant** is a mapping $\det: \text{Mat}(n; R) \rightarrow R$ given by: $\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \text{sgn}(\sigma) a_{1\sigma(1)} \dots a_{n\sigma(n)}$.
 This is called the **Leibniz formula**. For $n=0$ $\det(A) = 1$.
 Alternative characterisations are:
 • Let F be a field. The mapping $\det: \text{Mat}(n; F) \rightarrow F$ is the unique alternating multilinear form on n -tuples of column vectors with values in F that takes the value 1_F on the identity matrix.
 • Laplace Expansions: Let R be a comring and $A = (a_{ij}) \in \text{Mat}(n; R)$. For fixed i, j the i^{th} row/ j^{th} column expansion of the determinant is $\det(A) = \sum_{j=1}^n a_{ij} C_{ij} / \sum_{i=1}^n a_{ij} C_{ij}$.

Theorem: Calculations With The Determinant

• For R a comring, $A, B \in \text{Mat}(n; R)$:
 $-\det(AB) = \det(A)\det(B)$
 $-\det(A^T) = \det(A)$
 • For F a field, $A, B \in \text{Mat}(n; F)$:
 $-\det(A) \neq 0$ iff A is invertible
 $-$ If A is invertible then $\det(A^{-1}) = \det(A)^{-1}$
 $-\det(A^{-1}BA) = \det(B)$ (the det is independent of the choice of basis)
 • When performing Gaussian Elimination, row addition does not change the determinant, and row swaps change the sign only

Definition: Bi/Multilinear Forms

Let U, V, W be F -VSs. A **bilinear form** on $U \times V$ with values in W is a mapping $H: U \times V \rightarrow W$ linear in both entries. Ie, $\forall u_1, u_2 \in U, v_1, v_2 \in V$:
 • $H(\lambda u_1, v_1) = \lambda H(u_1, v_1)$ • $H(u_1 + u_2, v_1) = H(u_1, v_1) + H(u_2, v_1)$
 • $H(u_1, \lambda v_1) = \lambda H(u_1, v_1)$ • $H(u_1, v_1 + v_2) = H(u_1, v_1) + H(u_1, v_2)$
 For $U = V$, a bilinear form is **symmetric** if for all $u, v \in U$, $H(u, v) = H(v, u)$; and it is **alternating/antisymmetric** if for all $u \in U$, $H(u, u) = 0$.
 • If H is antisymmetric, $H(u, v) = -H(v, u)$.
 • Multilinear forms are just an extension of bilinear forms to higher dimensions - H must be linear in each argument.
 • A multilinear form is **alternating** if $\exists i \neq j$ with $v_i = v_j \implies H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = 0$ and if so, has the property $H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = -H(v_1, \dots, v_j, \dots, v_i, \dots, v_n)$ for all $v_1, \dots, v_n \in V$.
 • Thus for any $\sigma \in \mathfrak{S}_n$, $H(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \text{sgn}(\sigma) H(v_1, \dots, v_n)$

Theorem: Cramer's Rule and Jacobi

Cramer: For R be a comring, $A \in \text{Mat}(n; R)$; $A \cdot \text{adj } A = (\det(A)) I_n$
 • A is invertible iff $\det(A)$ is a unit in R , ie $\det(A) \in R^\times$
 Jacobi: For $A = (a_{ij})$, $a_{ij} = a_{ij}(t)$ func of t ; $\frac{d}{dt} \det(A) = \text{tr adj } A \frac{dA}{dt}$

Major Definition: Eigenvalues and Eigenvectors

Let $f: V \rightarrow V$ be an endo. of an F -VS V .
 $\lambda \in F$ is an **eigenvalue** of f iff there exists $n \neq 0$ $\vec{v} \in V$ s.t. $f(\vec{v}) = \lambda \vec{v}$.
 Each such $\vec{v} \in V$ is called an **eigenvector** of f with **eigenvalue** λ .
 $E(\lambda, f)$, the **eigspace** of f with **eigval** λ , is $\{\vec{v} \in V : f(\vec{v}) = \lambda \vec{v}\}$.
 • Each endo. of a n -z finite-dim VS over an algly-clsd field has an eigval.
 • For f with distinct eigvals $\lambda_1, \dots, \lambda_n$, the eigvecs $\vec{v}_1, \dots, \vec{v}_n$ are lin-ind.

Definition: The Characteristic Polynomial

Let R be a comring and $A \in \text{Mat}(n; R)$. The polynomial $\chi_A(x) := \det(xI_n - A) \in R[x]$ is called the **characteristic polynomial** of A .
 • For F a field, $A \in \text{Mat}(n; F)$; the eigvals of A are the roots of χ_A .

Theorem: Cayley-Hamilton Theorem

Let R be a comring and $A \in \text{Mat}(n; R)$.
 Then $\chi_A(x)$ evaluated at the matrix A , $\chi_A(A)$, equals zero.

Definition: Conjugacy

For R a comring, $A, B \in \text{Mat}(n; R)$.
 A and B are **conjugate** if $B = P^{-1}AP$ for invertible $P \in \text{GL}(n; R)$.
 • Conjugacy is an equivalence relation on $\text{Mat}(n; R)$.
 • The characteristic polynomials of conjugate matrices are the same

Definition: Triangularisability

Let $f: V \rightarrow V$ be an endo. of a finite-dim F -VS V . The following two statements are equivalent, and if they hold, f is **triangularisable**:
 (1) $\chi_f(x)$, decomposes into linear factors in $F[x]$.
 (2) V has an ordered basis $\mathcal{B} = (\vec{v}_1, \dots, \vec{v}_n)$ s.t. $f(\vec{v}_1) = a_{11}\vec{v}_1, f(\vec{v}_2) = a_{12}\vec{v}_1 + a_{22}\vec{v}_2, \dots, f(\vec{v}_n) = a_{1n}\vec{v}_1 + \dots + a_{nn}\vec{v}_n \in V$.
 Equivalently, the matrix $\mathcal{B}[f]_{\mathcal{B}}$ is upper triangular.
 • An endo. $A: F^n \rightarrow F^n$ is triangular iff A is conj to an upper triang mat B . Ie, any endo. of a finite-dim VS over an algly-clsd field is triangular.

Definition: Diagonalisability

• An endo. $f: V \rightarrow V$ of an F -VS is **diagonalisable** iff there exists a basis of V made up of eigvecs of f .
 If V is finite-dim, this is equivalent to: $\mathcal{O}$ ordered basis $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ such that $\mathcal{B}[f]_{\mathcal{B}} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Here $f(\vec{v}_i) = \lambda_i \vec{v}_i$.
 • A square matrix $A \in \text{Mat}(n; F)$ is **diagonalisable** iff the linear mapping $F^n \rightarrow F^n$ given by left multiplication by A is diagonalisable. That is, A is conjugate to a diagonal matrix $\{\exists P \in \text{GL}(n; F) \text{ s.t. } P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)\}$. In this case the columns of P are the vecs of a basis of F^n consisting of eigenvects of A with eigenvals $\lambda_1, \dots, \lambda_n$.

Inner Product Spaces

Major Definition: Inner Products And Their Spaces

Let V be an R -VS. An **inner product** on V is a mapping $(\cdot, \cdot): V \times V \rightarrow R$ that satisfies, for all $\vec{x}, \vec{y}, \vec{z} \in V$ and $\lambda, \mu \in R$:
 (1) $(\lambda \vec{x} + \mu \vec{y}, \vec{z}) = \lambda(\vec{x}, \vec{z}) + \mu(\vec{y}, \vec{z})$
 (2) $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$ (together with (1) implies symmetric bilinear form)
 (3) $(\vec{x}, \vec{x}) \geq 0$, = 0 iff $\vec{x} = 0$ (positive definite)
 • The **standard inner product** on $V = \mathbb{R}^n$ is the dot product
 • A **real inner product space** is a real VS endowed with an inner product
 • For V a VS over $\mathbb{C}$ the definition is similar, swap $\mathbb{R}$ for $\mathbb{C}$ everywhere; condition (2) reads $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$; and the standard inner product is $(\vec{v}, \vec{w}) = v_1 \overline{w_1} + \dots + v_n \overline{w_n}$. Note the SIP is not $\mathbb{C}$ -linear in the second variable.
 • In a $\mathbb{R}/\mathbb{C}$ -VS, the **length/inner product norm/norm** is $\|\vec{v}\| := \sqrt{(\vec{v}, \vec{v})}$.
 • Vectors of orthogonal length 1 are **units**, iff $(\vec{v}, \vec{w}) = 0$.
 • Sets S, T are orthogonal, $S \perp T$, if $\vec{u}, \vec{v} \in S \forall \vec{u} \in S, \vec{v} \in T$.
 • An **orthonormal family** of vecs in an IPS is a $(\vec{v}_i)_{i \in I}$ s.t. all $\vec{v}_i$ have $\|\vec{v}_i\| = 1$ and are pairwise orthogonal (by Kronecker delta, $(\vec{v}_i, \vec{v}_j) = \delta_{ij}$).
 • An orthonormal family that is a basis is an **orthonormal basis**. For V an IPS, $(\vec{v}_i)_{i \in I}$ an orthonormal basis; $\forall \vec{w} \in V, \vec{w} = \sum_{i \in I} (\vec{w}, \vec{v}_i) \vec{v}_i$.
 • Every finite-dim IPS has an orthonormal basis

Dfn: Orthogonal Complements/Projections

Let V be an IPS, $T \subseteq V$, and U be a finite-dim subspace of V .
 $T^\perp = \{\vec{v} \in V : \vec{v} \perp \vec{t} \forall \vec{t} \in T\}$ is the **orthogonal** to T .
 $T^\perp$ is always a subspace of V even if T is not, and $T^\perp = (T^\perp)^\perp$.
 The space $U^\perp$ is the **orthogonal complement** to U .
 • Every $\vec{v} \in V$ can be written as $\vec{v} = \vec{p} + \vec{q}$, $\vec{p} \in U$ and $\vec{q} \in U^\perp$.
 • U and $U^\perp$ are complementary, ie $V = U \oplus U^\perp$.
 The **orthogonal projection** from V onto U is $\pi_U: V \rightarrow V, \vec{v} = \vec{p} + \vec{q} \mapsto \vec{p}$.
 • π_U is a linmap with $\text{im}(\pi_U) = U$ and $\ker(\pi_U) = U^\perp$.
 • $\pi_U^2 = \pi_U$, ie π_U is idempotent
 • Let $\{\vec{v}_1, \dots, \vec{v}_n\}$ be an orthonormal basis of U .
 Then for all $\vec{v} \in V$, $\pi_U(\vec{v}) = \sum_{i=1}^n (\vec{v}, \vec{v}_i) \vec{v}_i$.

Major Theorem: The Cauchy-Schwarz Inequality

Let $\vec{v}, \vec{w}$ be vectors in an inner product space.
 Then $|\langle \vec{v}, \vec{w} \rangle| \leq \|\vec{v}\| \|\vec{w}\|$ with equality iff $\vec{v}, \vec{w}$ are linearly dependent.
 For V an IPS, $\vec{v}, \vec{w} \in V$, and λ a scalar (ie in the underpinning field):
 • $\|\vec{v}\| \geq 0$, = 0 iff $\vec{v} = \vec{0}$.
 • $\|\lambda \vec{v}\| = |\lambda| \|\vec{v}\|$
 • $\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$.
 • In a real IPS, $\exists$ unique $0 \leq \theta \leq \pi$ s.t. $\cos(\theta) = \frac{(\vec{v}, \vec{w})}{\|\vec{v}\| \|\vec{w}\|}$.

Theorem: Gram-Schmidt Process

Let $\vec{v}_1, \dots, \vec{v}_k$ be linearly ind. vectors in an IPS V . Then we can construct an orthonormal family $\vec{w}_1, \dots, \vec{w}_k$ such that for all $1 \leq i \leq k$, $\langle \vec{v}_1, \dots, \vec{v}_i \rangle = \langle \vec{w}_1, \dots, \vec{w}_i \rangle$. If V is finite-dim, we can use the G-S process to transform any basis into an orthonormal basis.
 • For $k=1$ take $\vec{w}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$ (normalise the length)
 • Then given $\vec{w}_1, \dots, \vec{w}_k$; $\vec{w}_{k+1} = \frac{\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i}{\|\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i\|}$

Definition: Adjoints and Self-Adjointness

Let V be an IPS. Two endo.s $T, S: V \rightarrow V$ are **adjoint** if for all $\vec{v}, \vec{w} \in V$, $(T\vec{v}, \vec{w}) = (S\vec{w}, \vec{v})$. In this case $S = T^*$ and T is the **adjoint** of T .
 • If V is finite-dim, IPS and $T: V \rightarrow V$ an endo; T^* exists and is unique.
 • An endo. of an IPS $T: V \rightarrow V$ is **self-adjoint** if $T = T^*$.
 • Let $T: V \rightarrow V$ be a self-adjoint linmap on an IPS V . Then:
 (1) Every eigenvalue of T is real
 (2) If λ, μ are distinct eigvals of T with eigenvects $\vec{v}, \vec{w}$, then $(\vec{v}, \vec{w}) = 0$
 (3) T has an eigenvalue.

Major Theorem: The Spectral Theorem

Let V be a finite-dim IPS and $T: V \rightarrow V$ be a self-adjoint linmap. Then V has an orthonormal basis consisting of eigenvectors of T .
 • Let $A \in \text{Mat}(n; \mathbb{R})$ be symmetric. Then $\exists$ orthogonal $P \in \text{Mat}(n; \mathbb{R})$ such that $P^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.
 • Let $A \in \text{Mat}(n; \mathbb{C})$ be hermitian. Then $\exists$ unitary $P \in \text{Mat}(n; \mathbb{C})$ s.t. $P^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.

Jordan Normal Form

Definition: Jordan Blocks

For $r \in \mathbb{N}$: The **nilpotent Jordan block** of size r is the $(r \times r)$ matrix $J(r)$ defined by the rule $J(r)_{ij} = 1$ for $j = i+1$ and = 0 otherwise. The **Jordan block** of size r and **eigval** λ , for $\lambda \in F$, is the $(r \times r)$ matrix $J(r, \lambda)$ defined by the rule $J(r, \lambda) = \lambda I_r + J(r)$ with $DN = ND$.

Major Theorem: Jordan Normal Form

Let F be an algebraically closed field, V a finite-dim VS, and $\phi: V \rightarrow V$ an endo, with characteristic polynomial $\chi_\phi(x) = (x - \lambda_1)^{a_1} \dots (x - \lambda_s)^{a_s}$.
 Then $\exists$ an ordered basis $\mathcal{B}$ of V s.t. the matrix of ϕ w.r.t. $\mathcal{B}$ is block diag with Jordan blocks on the diag. That is, $\mathcal{B}[\phi]_{\mathcal{B}} = \text{diag}(J(r_{11}, \lambda_1), \dots, J(r_{1m_1}, \lambda_1), J(r_{21}, \lambda_2), \dots, J(r_{sm_s}, \lambda_s))$ with $r_{11}, \dots, r_{1m_1}, r_{21}, \dots, r_{sm_s} \geq 1$ s.t. $a_i = r_{i1} + r_{i2} + \dots + r_{im_i}$, $(1 \leq i \leq s)$.